A Phase Vocoder Approach To Pitch Shifting

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1 Introduction

The manipulation of pitch in digital signals is not achieved by simply scaling frequencies, but through structured transformations in the time-frequency domain.

2 Background

2.1 Discrete Fourier Transform

The Discrete Fourier Transform is a fundamental tool used in digital signal processing that enables sophisticated analysis and synthesis of discrete-time signals [6]. The DFT transforms a signal from the time domain into its frequency-domain representation, allowing the signal to be expressed as a sum of sinusoids [6]. Mathematically, the DFT is defined as:

$$X[m] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi n \frac{m}{N}} \tag{1}$$

where N is the number of samples from the input signal $x[n]$, n denotes the time-domain index, m represents the frequency bin index, n is the index of the input sequenc in the time-domain and $j = \sqrt{-1}$. Each coeffecient in $X[m]$ is complex-valued and can be expressed in terms of magnitude and phase, which describe the contribution of a corresponding frequency component to the overall signal [7].

2.2 Windowing

Windowing is a function used to isolate parts of a discrete time signal so that analysis on the signal can be performed on contiguous blocks [1]. More specifically, the input signal is multiplied by a windowing function whose domain is nonzero for a particular range. This results in a small localized snapshot of the input signal, and every other value outside of the windows domain is reduced to zero [9]. Figure 1 below demonstrates the functionality of a window function.

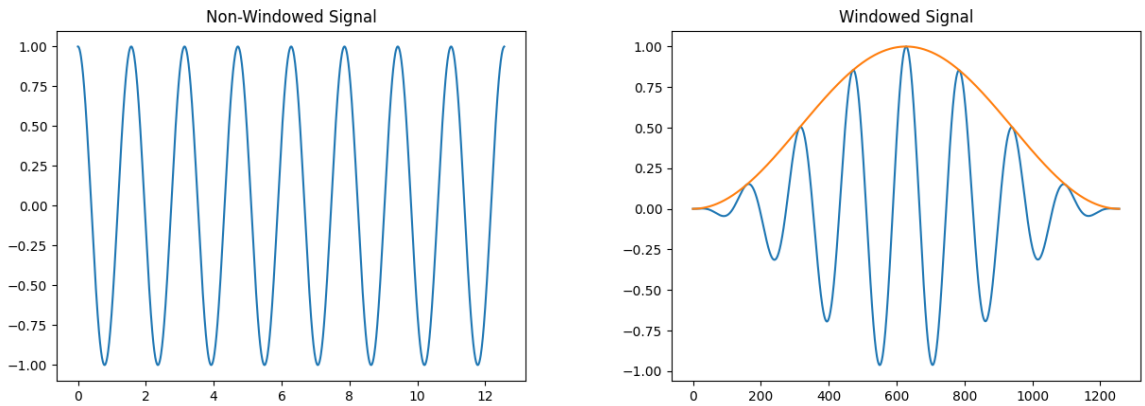

Figure 1: After applying a window function (highlighted by the orange cosine curve), the signal is tapered at the edges, resulting in a temporally localized signal.

As dipicted by Figure 1, after applying the window function to the example signal, the edges of the signal tend toward zero. In this example, the Hanning window function was used to shape the signal, however, there are multiple windowing functions such as the Rectangle window function, Triangle window function and Hamming window function [7]

2.3 Short-Time Fourier Transform

The Short-Time Fourier Transform (STFT) is a time-frequency representation used to compute the DFT of contiguous windowed segments of a signal. This is accomplished by multiplying the signal by a windowing function so that only a small portion of the signal is nonzero [2]. Mathematically, it is similar to equation (1) but with a few modifications: the addition of a windowing function, the domain of the input signal, and how the output is interpreted. Below is the STFT:

$$X[m,k] = \sum_{n=0}^{N-1} x[n+kH]w[n]e^{-2\pi jmn/N} \tag{2}$$

Where $X[m,k]$ is the 2-dimensional output array indexed by both frequency bin and time frame and $x[n+kH]$ is the section of the input signal determined by frame index and hop size [9]. The outcome of the STFT is a representation that transforms a signal from the time domain to the frequency domain. In this form, the signal's frequency components can be systematically modified and recombined to reconstruct a new signal.

3 Related Works

3.1 Classical Phase Vocoder Approaches

Pitch-shifting has been approached by a number of time–frequency methods. Early methods, such as Dolson's (1986), employed the use of the Short-Time Fourier Transform in conjunction with the phase vocoder algorithm. In particular, Dolson describes a pitch-shifting method that relies on time-scale manipulation to modify the pitch of a given signal [2]. For example, by stretching a signal, making it longer in the time domain, and then rescaling it back to its original length, the overall frequency content of the signal is effectively decreased, which corresponds to lowering the signal's pitch. Dolson acknowledges, however, that this approach is imperfect: specific constraints must be enforced when performing a pitch shift, as failing to do so may introduce unwanted artifacts into the signal or result in a reconstructed signal that has lost information [2]. While effective for monophonic (single note) pitch scaling, this classical time-scale modification framework treats spectral bins independently, which can lead to phase incoherence and unwanted artifacts. These limitations motivated subsequent refinements to the phase vocoder algorithm.

3.2 Phase Vocoder Optimizations

Unlike the original process chain described by Dolson (1986), which applies a uniform scaling factor to all spectral components, later work by Laroche and Dolson (1999) introduces refinements that improve frequency coherence [4]. In particular, they highlight a new peak-detection paradigm, where peaks refer to how much a particular frequency is in the input signal. Once the peaks are detected, they can be shifted around by a pitch-factor and unlike the version described by Dolson (1986), the pitch-factor is not restricted to an integer amount [4].

In addition to optimizing frequency resolution, modern implementations of the phase vocoder shift away from the standard two-step timescale and resample approach and adopt an analysis and resynthesis paradigm [3]. The new approach leverages phase information obtained from the STFT step to more accurately reconstruct the modified signal [3]. More specifically, a phase adjustment value can be derived from the frequency bins across two frames and when added to the original frequency approximation the accuracy of the frequency estimation is increased [5].

3.3 Alternative Pitch-Shifting Methods

The prior sections highlight the phase vocoder approach to pitch-shifting, a frequency-domain algorithm that converts a signal from the time domain to the frequency domain to compute the pitch shift. An alternative approach uses time-scale modification (TSM) algorithms, which operate entirely in the time domain [8]. Since TSM algorithms are more naturally suited to speed changes, they typically require a resample step to restore the modified signal to its original duration [8]. These algorithms must incorporate a resmaple step inorder to convert the modified signal back to the original signal's duration [8].

3.4 Positioning of the Present Work

4 Methodology

Dijkstra's algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [**?**].

5 Results

An exemplary figure is shown in Figure **??**. Any time a figure is used, it must have a capation and be directly referred to and discussed in the text.

6 Conclusion

References

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